類別資料分析 第3章 (1)

The Analysis of Categorical Data

Chapter 3 (1) : Three-way Contingency Tables

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Three-way Contingency Tables

*conditional X-Y association, conditioned on Z=k

$\theta_{XY(k)} = \frac{\mu_{11k}\,\mu_{22k}}{\mu_{12k}\,\mu_{21k}}$: the X-Y conditional odds ratio $\left(\hat{\theta}_{XY(k)} = \frac{n_{11k}\,n_{22k}}{n_{12k}\,n_{21k}}\right)$

*The X-Y marginal odds ratio is $\theta_{XY} = \frac{\mu_{11+}\,\mu_{22+}}{\mu_{12+}\,\mu_{21+}}$ $\left(\hat{\theta}_{XY} = \frac{n_{11+}\,n_{22+}}{n_{12+}\,n_{21+}}\right)$

Collapse over Z

Z=1		Y	
		1	2
X	1	μ_{111}	μ_{121}
	2	μ_{211}	μ_{221}

Partial table

Z=2		Y	
		1	2
X	1	μ_{112}	μ_{122}
	2	μ_{212}	μ_{222}

Partial table

合併 →

		Y	
		1	2
X	1	μ_{11+}	μ_{12+}
	2	μ_{21+}	μ_{22+}

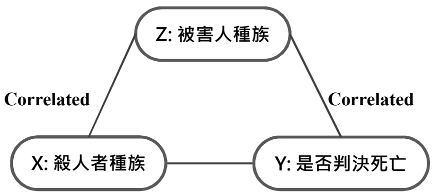
X-Y marginal table

Three-way Contingency Tables

*Simpson's paradox

Z : Victims' race = $\begin{cases} W, white \\ B, black \end{cases}$;

X : Defendants' race = $\begin{cases} W, white \\ B, black \end{cases}$; Y : Death penalty



Z=W		Y	
		Yes	No
X	W	53	414
	B	11	37

Z=B		Y	
		Yes	No
X	W	0	16
	B	4	139

對 Z 合併 →

		Y	
		Yes	No
X	W	53	430
	B	15	176

$\hat{\theta}_{XY(1)} = \frac{53\times 37}{11\times 414} \approx 0.43 < 1$

負相關

$\hat{\theta}_{XY(2)} = \frac{0.5\times 139.5}{4.5\times 16.5} \approx 0.94 < 1$

負相關

$\hat{\theta}_{XY} = \frac{53\times 176}{15\times 430} \approx 1.45 > 1$

正相關

Three-way Contingency Tables

*Simpson's paradox

對 Y 合併		X	
		W	B
Z	W	467	48
	B	16	143

$\hat{\theta}_{XZ} = \frac{467\times 143}{16\times 48} \approx 87$（大部分白人都是被白人殺害，

黑人都是被黑人殺害。）

對 X 合併		Y	
		Yes	No
X	W	64	451
	B	4	155

$\hat{\theta}_{YZ} = \frac{64\times 155}{4\times 451} \approx 5.5$（被害人是白人時，殺人者被判死刑的勝算是

被害人是黑人時的 5.5 倍。）

Three-way Contingency Tables

*Simpson’s paradox

- 由於被害人種族 (Z) 與是否判決死亡 (Y) 有關，且被害人種族與殺人者種族 (X) 有關，即被害人種族為混淆因子 (Confounding factor)，故[被害人種族]干擾[殺人者種族]與[是否判決死亡]之間的相關性（因此只看 Marginal table，容易誤導結論。）
- 合併前 Partial table 的 cell count 越大，對合併後 Marginal table的影響越大。

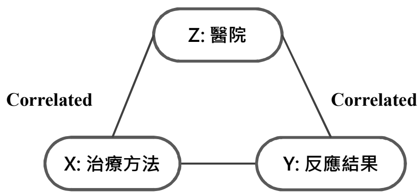
Three-way Contingency Tables

*Simpson’s paradox

Z : Clinic $(1, 2)$;

X : Treatment (A, B) ;

Y : Response (Success, Failure)



Z=1		Y	
		S	F
X	A	18	12
	B	12	8

$\hat{\theta}_{XY(1)} = \frac{18\times8}{12\times12} = 1$

無關

Z=2		Y	
		S	F
X	A	2	8
	B	8	32

$\hat{\theta}_{XY(2)} = \frac{2\times32}{8\times8} = 1$

無關

對 Z 合併 →

		Y	
		Yes	No
X	W	20	20
	B	20	40

$\hat{\theta}_{XY} = \frac{20\times40}{20\times20} = 2 > 1$

正相關

Three-way Contingency Tables

*Simpson's paradox

對 Y 合併		Z	
		1	2
X	A	30	10
	B	20	40

$\hat{\theta}_{XZ} = \frac{30\times40}{20\times10} = 6$（醫院 1 的實驗者多數吃 A 藥，

醫院 2 的實驗者多數吃 B 藥。）

對 X 合併		Z	
		1	2
Y	S	30	10
	F	20	40

$\hat{\theta}_{YZ} = \frac{30\times40}{20\times10}$ =6（醫院 1 的實驗結果多數為成功，

醫院 2 的實驗結果多數為失敗。）

Three-way Contingency Tables

*Simpson's paradox

- 由於反應結果是否成功 (Y) 與醫院 (Z) 有關，治療方法 (X) 也和醫院有關，即醫院為混淆因子，干擾治療方法與反應結果的相關性。
- 針對多因子列連表，應避免合併任意因子，以免誤導結果。

Three-way Contingency Tables

*有 k 個 2×2 tables

(1) k 個 tables 中的相關性是否一致？ (i.e. $H_0 : \theta_{XY(1)} = \theta_{XY(2)} = \cdots = \theta_{XY(k)}$)

· Breslow-Day Test for Homogeneity of Odds Ratios

1. $H_0 : \theta_{XY(1)} = \theta_{XY(2)} = \cdots = \theta_{XY(k)}$ vs. H_1 : at least one $\theta_{XY(i)}$ is unequal

2. Test statistic : $X^2 = \sum_k \frac{n_{11k} - \hat{\mu}_{11k}}{\widehat{Var}(n_{11k})} \sim X^2_{k-1}$ if $\hat{\mu}_{ijk} \geq 5$ in at least about 80% of the cells,

where $\mu_{11k} = E(n_{11k} | n_{1+k}, \theta_{XY(k)} = \theta, \forall k)$

$\hat{\theta}_{MH} = \frac{\sum_k n_{11k} n_{22k}/n_{++k}}{\sum_k n_{12k} n_{21k}/n_{++k}}$: the common odds ratio estimator (or Mantel-Haenszel estimator)

solve $\hat{\mu}_{11k}$ s.t. $\frac{\hat{\mu}_{11k}(n_{+2k} - n_{1+k} + \hat{\mu}_{11k})}{(n_{1+k} - \hat{\mu}_{11k})(n_{+1k} - \hat{\mu}_{11k})} = \hat{\theta}_{MH}$

Three-way Contingency Tables

(1)有 k 個 2×2 tables, k 個 tables 中的相關性是否一致？

· Breslow-Day Test for Homogeneity of Odds Ratios

1. $H_0 : \theta_{XY(1)} = \theta_{XY(2)} = \cdots = \theta_{XY(k)}$

2. Test statistic : $X^2 = \sum_k \frac{n_{11k} - \hat{\mu}_{11k}}{\widehat{Var}(n_{11k})}$ ($\because$ $n_{11k} \sim$ generalized hypergeometric with parameter $\hat{\theta}$

$$\therefore \widehat{Var}(n_{11k}) = \left(\frac{1}{\hat{\mu}_{11k}} + \frac{1}{n_{1+k} - \hat{\mu}_{11k}} + \frac{1}{n_{+1k} - \hat{\mu}_{11k}} + \frac{1}{n_{+2k} - n_{1+k} + \hat{\mu}_{11k}}\right)^{-1}$$

3. Reject H_0 if $X^2 > X^2_{k-1,\alpha}$

Reject H_0 代表 3 個類別變數 X, Y, Z 彼此有交互作用

Do not reject H_0 代表 k 個 2×2 tables 的相關性一致

expected table

μ_{11k}	$n_{1+k}-\mu_{11k}$	n_{1+k}
$n_{+1k}-\mu_{11k}$	$n_{+2k}+\mu_{11k}$	n_{2+k}
n_{+1k}	n_{+2k}	

Three-way Contingency Tables

*有 k 個 2×2 tables

(2) 若 k 個 tables 中相關性一致，則共同的相關性是否顯著？ (i. e. H_0：$\theta_{XY(i)} = 1, i = 1, 2, \ldots, k$)

‧ Cochran-Mantel-Haenszel Test (CMH Test)

1. H_0：$\theta_{XY(1)} = \theta_{XY(2)} = \cdots = \theta_{XY(k)} = 1$ vs. H_1：at least one table with $\theta_{XY(i)} \neq 1$

2. Test statistic : $CMH = \frac{[\sum_k (n_{11k} - \hat{\mu}_{11k})]^2}{\sum_k \widehat{Var}(n_{11k})} \sim X_1^2$ for large n,

   where $\hat{\mu}_{11k} = \hat{E}(n_{11k}) = \frac{n_{1+k}\, n_{+1k}}{n_{++k}}$, $\widehat{Var}(n_{11k}) = \frac{n_{1+k}\, n_{+1k}\, n_{2+k}\, n_{+2k}}{n_{++k}^2 (n_{++k} - 1)}$

3. Reject H_0 if $X^2 > X^2_{k-1,\alpha}$

Z=k		Y: 1	Y: 2	
X	1	n_{11k}	n_{12k}	n_{1+k}
	2	n_{21k}	n_{22k}	n_{2+k}
		n_{+1k}	n_{+2k}	n_{++k}

缺點：當有些 partial tables 是正相關，有些是負相關時，CMH的分子會因正負抵消掉而變小，

故此檢定只適用於每一 strata 的 partial table 相關性差不多時。

Three-way Contingency Tables

*有 k 個 2×2 tables

(3) 若顯著，如何估計共同的Odds Ratio？

‧ Estimation of Common Odds Ratio

(i) Case-Control (for odds ratio)

$$\hat{\theta}_{MH} = \frac{\sum_k n_{11k}\, n_{22k}/n_{++k}}{\sum_k n_{12k}\, n_{21k}/n_{++k}} \ ; \hat{\theta}_k = \frac{n_{11k}\, n_{22k}/n_{++k}}{n_{12k}\, n_{21k}/n_{++k}} \ ;$$

$$\hat{\theta}_{logit} = \exp\left[\frac{\sum_k \omega_k \ln \hat{\theta}_k}{\sum_k \omega_k}\right], \text{where } \omega_k = \left[Var(\ln \hat{\theta}_k)\right]^{-1}$$

Three-way Contingency Tables

*有 k 個 2 × 2 tables

(3) 若顯著，如何估計共同的相關性？

· Estimation of Common Odds Ratio

(ii) Cohort (column 1 risk)

$$\widehat{RR}^{(1)} = \frac{\sum_k n_{11k}\, n_{2+k}/n_{++k}}{\sum_k n_{21k}\, n_{1+k}/n_{++k}}\ ; \widehat{RR}_k^{(1)} = \frac{n_{11k}/n_{1+k}}{n_{21k}/n_{2+k}} = \frac{n_{11k}\, n_{2+k}}{n_{21k}\, n_{1+k}}\ ;$$

$$\widehat{RR}_{logit} = \exp\left[\frac{\sum_k \omega_k \ln \widehat{RR}_k^{(1)}}{\sum_k \omega_k}\right], \text{where } \omega_k = \left[Var(\ln \widehat{RR}_k^{(1)})\right]^{-1}$$

(iii) Cohort (column 2 risk)

Three-way Contingency Tables

*有 k 個 2 × 2 tables

(3) 若顯著，如何估計共同的相關性？

· Estimation of Common Odds Ratio

(iii) Cohort (column 2 risk)

$$\widehat{RR}^{(2)} = \frac{\sum_k n_{12k}\, n_{2+k}/n_{++k}}{\sum_k n_{22k}\, n_{1+k}/n_{++k}}\ ; \widehat{RR}_k^{(2)} = \frac{n_{12k}/n_{1+k}}{n_{22k}/n_{2+k}} = \frac{n_{12k}\, n_{2+k}}{n_{22k}\, n_{1+k}}\ ;$$

$$\widehat{RR}_{logit} = \exp\left[\frac{\sum_k \omega_k \ln \widehat{RR}_k^{(2)}}{\sum_k \omega_k}\right], \text{where } \omega_k = \left[Var(\ln \widehat{RR}_k^{(2)})\right]^{-1}$$

Three-way Contingency Tables

e. g. Chinese Smoking and Lung Cancer Study

$$\cdot\ \mathbf{CMH} = \frac{[\sum_k (n_{11k} - \hat{\mu}_{11k})]^2}{\sum_k \widehat{\mathrm{Var}}(n_{11k})} = \frac{(2930-2562.5)^2}{482.1} \approx 280.1$$

· **Common odds ratio estimator**

$$\hat{\theta}_{MH} = \frac{\sum_k n_{11k}\, n_{22k}/n_{++k}}{\sum_k n_{12k}\, n_{21k}/n_{++k}} = \frac{\left(\frac{126\times61}{322}\right)+\cdots+\left(\frac{104\times36}{250}\right)}{\left(\frac{35\times100}{322}\right)+\cdots+\left(\frac{21\times89}{250}\right)} \approx 2.17$$

City	Smoking	Lung Cancer Yes	Lung Cancer No	Odds Ratio	n_{11k}	$\hat{\mu}_{11k}$	$\widehat{\mathrm{Var}}(n_{11k})$
Beijing	Smokers	126	100	2.20	126	113.0	16.9
	Nonsmokers	35	61				
Shanghai	Smokers	908	688	2.14	908	773.2	179.3
	Nonsmokers	497	807				
Shenyang	Smokers	913	747	2.18	913	799.3	143.3
	Nonsmokers	336	598				
Nanjing	Smokers	235	172	2.85	235	203.5	31.1
	Nonsmokers	58	121				
Harbin	Smokers	402	308	2.32	402	355.0	57.1
	Nonsmokers	121	215				
Zhengzhou	Smokers	182	156	1.59	182	169.0	28.3
	Nonsmokers	72	98				
Taiyuan	Smokers	60	99	2.37	60	53.0	9.0
	Nonsmokers	11	43				
Nanchang	Smokers	104	89	2.00	104	96.5	11.0
	Nonsmokers	21	36				
					2930	2562.5	482.1

Three-way Contingency Tables

e. g. Chinese Smoking and Lung Cancer Study

$$\log\hat{\theta}_{MH} = \log 2.17 \approx 0.777;\ \widehat{ASE}(\log\hat{\theta}_{MH}) = 0.046$$

$$\therefore \text{ A 95\% C.I. for } \log\theta \text{ is } 0.777 \pm 1.96\cdot 0.046$$

$$\approx (0.686, 0.868)$$

$$\Rightarrow \text{An approximate 95\% C.I. for } \theta \text{ is } (e^{0.686}, e^{0.868})$$

$$\approx (1.98, 2.38)$$

$$\cdot\ \text{Breslow} - \text{Day Statistic} = 5.2,\ \text{df} = 7,$$

$$\text{p} - \text{value} = 0.64$$

City	Smoking	Lung Cancer Yes	Lung Cancer No	Odds Ratio	n_{11k}	$\hat{\mu}_{11k}$	$\widehat{\mathrm{Var}}(n_{11k})$
Beijing	Smokers	126	100	2.20	126	113.0	16.9
	Nonsmokers	35	61				
Shanghai	Smokers	908	688	2.14	908	773.2	179.3
	Nonsmokers	497	807				
Shenyang	Smokers	913	747	2.18	913	799.3	143.3
	Nonsmokers	336	598				
Nanjing	Smokers	235	172	2.85	235	203.5	31.1
	Nonsmokers	58	121				
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	Nonsmokers	121	215				
Zhengzhou	Smokers	182	156	1.59	182	169.0	28.3
	Nonsmokers	72	98				
Taiyuan	Smokers	60	99	2.37	60	53.0	9.0
	Nonsmokers	11	43				
Nanchang	Smokers	104	89	2.00	104	96.5	11.0
	Nonsmokers	21	36				
					2930	2562.5	482.1

Three-way Contingency Tables

***SAS code**

```
options nodate nonotes ps=60;
data cmh;
 input center smoke cancer count @@;
cards;
1 1 1 126 1 1 2 100 1 2 1  35 1 2 2  61
2 1 1 908 2 1 2 688 2 2 1 497 2 2 2 807
3 1 1 913 3 1 2 747 3 2 1 336 3 2 2 598
4 1 1 235 4 1 2 172 4 2 1  58 4 2 2 121
5 1 1 402 5 1 2 308 5 2 1 121 5 2 2 215
6 1 1 182 6 1 2 156 6 2 1  72 6 2 2  98
7 1 1  60 7 1 2  99 7 2 1  11 7 2 2  43
8 1 1 104 8 1 2  89 8 2 1  21 8 2 2  36
;
proc freq;
  weight count;
  tables center*smoke*cancer/nocol norow nopercent cmh chisq;
run;
```

Three-way Contingency Tables

***SAS output**

```
      TABLE 8 OF SMOKE BY CANCER
       CONTROLLING FOR CENTER=8

SMOKE      CANCER
Frequency|        1|        2|  Total
---------+---------+---------+
       1 |    104 |      89 |    193
---------+---------+---------+
       2 |     21 |      36 |     57
---------+---------+---------+
Total          125       125      250
```

```
Statistic                         DF      Value        Prob
-----------------------------------------------------------
Chi-Square                         1      5.113       0.024
Likelihood Ratio Chi-Square        1      5.161       0.023
Continuity Adj. Chi-Square         1      4.454       0.035
Mantel-Haenszel Chi-Square         1      5.093       0.024
Fisher's Exact Test (Left)                            0.992
                    (Right)                        1.71E-02
                    (2-Tail)                       3.43E-02
Phi Coefficient                           0.143
Contingency Coefficient                   0.142
Cramer's V                                0.143
```

Three-way Contingency Tables

*SAS output

```
              SUMMARY STATISTICS FOR SMOKE BY CANCER
                     CONTROLLING FOR CENTER

     Cochran-Mantel-Haenszel Statistics (Based on Table Scores)

   Statistic    Alternative Hypothesis    DF       Value      Prob
   ------------------------------------------------------------------
      1         Nonzero Correlation        1      280.138     0.000

        Estimates of the Common Relative Risk (Row1/Row2)
                                                          95%
   Type of Study      Method                 Value    Confidence Bounds
   ------------------------------------------------------------------
   Case-Control       Mantel-Haenszel        2.174      1.985     2.382
     (Odds Ratio)     Logit                  2.173      1.983     2.382
   Cohort             Mantel-Haenszel        1.519      1.447     1.595
     (Col1 Risk)      Logit                  1.513      1.436     1.594
   Cohort             Mantel-Haenszel        0.700      0.671     0.730
     (Col2 Risk)      Logit                  0.701      0.673     0.730

        Breslow-Day Test for Homogeneity of the Odds Ratios

 Chi-Square = 5.200    DF = 7     Prob = 0.636   Total Sample Size = 8419
```

Three-way Contingency Tables

***補充：Measure of Association for Ordinal Data (Based on concordant and discordant pairs)**

考慮 bivariate data (x_i, y_i)，對任意兩個觀測值 (x_i, y_i) 和 (x_j, y_j)，若 $x_i > x_j$ & $y_i > y_j$ 或 $x_i < x_j$ & $y_i < y_j$，則稱此兩觀測值為一致性配對(Concordant pairs)，否則為非一致性配對(Discordant pairs) ，此種觀念經常用在 nonparametric 中之 measure of rank correlation 中出現。例如下列 bivariate data (measure of aggressive) :

Twin set	1	2	3	4	5	6	7	8	9	10	11	12
First born	86	71	77	68	91	72	77	91	70	71	88	87
Second born	88	77	76	64	96	72	65	90	65	80	81	72

Three-way Contingency Tables

*Measure of Association for Ordinal Data (Based on concordant and discordant pairs)

	(x_i, y_i)	Concordant Pairs Below (x_i, y_i)	Discordant Pairs Below (x_i, y_i)
	(68,64)	11	0
	(70,65)	9	0
tie	(71,77)	4	4
	(71,80)	4	4
	(72,72)	5	1
tie	(77,65)	5	0
	(77,76)	4	1
	(86,88)	2	2
	(87,72)	3	0
	(88,81)	2	0
tie	(91,90)	0	0
	(91,96)	0	0
		$N_c = 49$	$N_d = 12$

$$\text{Kendall's } \tau = \frac{N_c - N_d}{n(n-1)/2} = \frac{49-12}{12 \cdot 11/2} \approx 0.5606$$

∴ There is a positive rank correlation between aggression scores in the twins.

（i. e. 若雙胞胎中，老大攻擊性強，則老二的攻擊性也強。）

/ 同樣的觀念可應用在 Contingency table，例如下頁 data。

Three-way Contingency Tables

e. g. 教育程度越高者是否越贊成（或反對）墮胎的合法化？

	反對	無意見	贊成	
高中以下	209	101	237	547
高中	151	126	426	703
高中以上	16	21	138	175
	376	248	801	1425

209		
	126	426
	21	138

$209 \cdot (126 + 426 + 21 + 138)$

	101	
		426
		138

$101 \cdot (426 + 138)$

151		
	21	138

$151 \cdot (21 + 138)$

	126	
		138

$126 \cdot 138$

Three-way Contingency Tables

e. g. 教育程度越高者是否越贊成（或反對）墮胎的合法化？

· $\mathbf{Concordant\ pairs = C = 209 \cdot (126 + 426 + 21 + 138) + 101 \cdot (426 + 138) + 151 \cdot (21 + 138)}$

$\mathbf{+126 \cdot 138 = 246960}$

· $\mathbf{Discordant\ pairs = D = 237 \cdot (151 + 126 + 16 + 21) + 101 \cdot (151 + 16) + 426 \cdot (16 + 21)}$

$\mathbf{+126 \cdot 16 = 109063}$

$\because \mathbf{C > D}$ $\therefore$ 教育程度越高者越贊成墮胎合法化

Three-way Contingency Tables

➢ 一般而言，$\begin{Bmatrix} \mathbf{C} = \sum_{i<k} \sum_{j<\ell} x_{ij}\, x_{k\ell} \\ \mathbf{D} = \sum_{i<k} \sum_{j>\ell} x_{ij}\, x_{k\ell} \end{Bmatrix}$ $: \mathbf{C + D} = \binom{n}{2}$ 對連續資料

- **Total number of pairs tied on the row :** $\mathbf{T_x} = \sum_i \binom{x_{i+}}{2}$
- **Total number of pairs tied on the column :** $\mathbf{T_y} = \sum_j \binom{x_{+j}}{2}$
- **Total number of pairs tied on the X & Y :** $\boldsymbol{T_{xy}} = \sum_i \sum_j \binom{x_{ij}}{2}$

$\Rightarrow \binom{n}{2} = \mathbf{C} + \mathbf{D} + \boldsymbol{T}_x + \mathbf{T}_y - \mathbf{T}_{xy}$

- 若 $\mathbf{C > D}$ 表示 **X** 大，**Y** 隨之大；**X** 小，**Y** 也隨之小。

Three-way Contingency Tables

*Goodman-Kruskal's gamma

$\text{gamma} = \hat{\gamma} = \frac{C-D}{C+D} = \frac{C}{C+D} - \frac{D}{C+D}, -1 \le \hat{\gamma} \le 1$

proportion of concordant pairs : $\frac{C}{C+D}$

proportion of discordant pairs : $\frac{D}{C+D}$

- if $\pi_D = 0 \Rightarrow \gamma = 1$: 正相關
- if $\pi_C = 0 \Rightarrow \gamma = -1$: 負相關
- if $R \perp C \Rightarrow \gamma = 0$: 獨立

- Population version : $\gamma = \frac{\pi_C - \pi_D}{\pi_C + \pi_D}$, where $\pi_C = \sum_{i<k}\sum_{j<\ell} P_{ij}\, P_{k\ell}$, $\pi_D = \sum_{i<k}\sum_{j>\ell} P_{ij}\, P_{k\ell}$

- For 2×2 table, $\gamma = \frac{P_{11}P_{22} - P_{12}P_{21}}{P_{11}P_{22} + P_{12}P_{21}} = \frac{OR-1}{OR+1} = \text{yule's } Q$, where $OR = \frac{P_{11}P_{22}}{P_{12}P_{22}}$

Three-way Contingency Tables

*Somer's D (C*R)

$\widehat{D}(C|R) = \frac{C-D}{\binom{n}{2} - T_X}$ (As column variable is a response variable.)

$D(C|R) = \frac{\pi_C - \pi_D}{1 - \sum_i P_{i+}^2}$

- For 2×2 table, $D(C|R) = \frac{P_{11}}{P_{1+}} - \frac{P_{21}}{P_{2+}}$: difference of proportion

*Kendall's tau

$\hat{\tau} = \frac{C-D}{\binom{n}{2}}$: 適用於連續資料 ($T_X = 0$ & $T_Y = 0 \therefore C + D = \binom{n}{2}$)

Three-way Contingency Tables

***Kendall's tau b**

$$\hat{\tau}_b = \frac{C-D}{\sqrt{\left[\binom{n}{2}-T_X\right]\left[\binom{n}{2}-T_Y\right]}}$$

$$\tau_b = \frac{\pi_C-\pi_D}{\sqrt{\left[1-\sum_i P_{i+}^2\right]\left[1-\sum_j P_{+j}^2\right]}}$$

$$\hat{\tau}_b^2 = \widehat{D}(C|R)\cdot\widehat{D}(R*C)$$

· **For 2×2 table，$\hat{\tau}_b$ 為一 correlation coefficient，解釋如下：**

For each pair (x_a, y_a), (x_b, y_b), define

$$\mathrm{sign}(x_a-x_b)=U_{ab}=\begin{Bmatrix}-1, x_a<x_b\\ 0, x_a=x_b\\ 1, x_a>x_b\end{Bmatrix} \text{ and } \mathrm{sign}(y_a-y_b)=V_{ab}=\begin{Bmatrix}-1, y_a<y_b\\ 0, y_a=y_b\\ 1, y_a>y_b\end{Bmatrix}$$

Three-way Contingency Tables

***Kendall's tau b**

· $U_{ab}=-U_{ba}$; $V_{ab}=-V_{ba}$

· $U_{ab}V_{ab}=\begin{cases}1, \text{for concordant pair}\\ -1, \text{for discordant pair}\end{cases}$

· $U_{ab}^2=\begin{Bmatrix}1, \text{if } x_a\neq x_b\\ 0, \text{if } x_a=x_b\end{Bmatrix}$; $V_{ab}^2=\begin{Bmatrix}1, \text{if } y_a\neq y_b\\ 0, \text{if } y_a=y_b\end{Bmatrix}$

· For $\binom{n}{2}$ ordered pairs, $\sum\sum_{a\neq b}U_{ab}V_{ab}=2(C-D)$

$$\sum\sum_{a\neq b}U_{ab}^2=2\left[\binom{n}{2}-T_X\right];\ \sum_{a\neq b}V_{ab}^2=2\left[\binom{n}{2}-T_Y\right]$$

$$\sum\sum_{a\neq b}U_{ab}=\sum\sum_{a\neq b}V_{ab}=0\ (\because U_{ab}=-U_{ba};\ V_{ab}=-V_{ba})$$

$$\therefore \frac{\sum\sum_{a\neq b}U_{ab}V_{ab}}{\sqrt{(\sum\sum_{a\neq b}U_{ab}^2)(\sum_{a\neq b}V_{ab}^2)}}\equiv\frac{C-D}{\sqrt{\left[\binom{n}{2}-T_X\right]\left[\binom{n}{2}-T_Y\right]}}=\hat{\tau}_b$$

：相當於 U_{ab} 和 V_{ab} 的樣本相關係數

Three-way Contingency Tables

*$R \times C$ tables for ordinal-nominal data

假如 $R \times C$ table 中，其中一個因子有順序關係 (ordered, ordinal)，若使用一般 χ^2 test 時，power 會非常低，此時，可改用特別的對立假設（如：假設機率有線性趨勢）。例如：

	X: Age				
	6-7	8-9	10-11	12-13	14-15
Lie	6	18	19	27	25
Do not lie	15	31	31	31	19
Total	21	49	50	59	44
Score Z	-2	-1	0	1	2
$\hat{P}_Z$	0.286	0.367	0.38	0.458	0.568

where $P_Z = P(Y = \text{Lie}|Z)$：年齡為 Z 說謊之機率

$\chi^2 = 6.809$, p-value=0.146

Three-way Contingency Tables

*$R \times C$ tables for ordinal-nominal data

Fit linear model $P_j = \beta_0 + \beta_1 Z_j$

$H_0: \beta_1 = 0 \Leftrightarrow H_0: P_j = P, \forall j$

令 $\hat{P}_j = \frac{n_{1j}}{n_{+j}} \Rightarrow \widehat{Var}(\hat{P}_j) = \frac{\bar{P}(1-\bar{P})}{n_{+j}}, \bar{P} = \frac{\sum n_{1j}}{\sum n_{+j}}$ under H_0

Regress $\hat{P}_j$ on Z_j by weighted-least-square，則 $\hat{\beta}_1 = \frac{\sum_j (n_{1j} - n_{+j}\bar{P})(Z_j - \bar{Z})}{\sum_j (Z_j - \bar{Z})^2 \cdot n_{+j}}$,

where $\bar{Z} = \frac{\sum n_{+j} Z_j}{\sum n_{+j}}, Var(\hat{\beta}_1) = \frac{\bar{P}(1-\bar{P})}{\sum_j (Z_j - \bar{Z})^2 \cdot n_{+j}}$

由上述 data 得 $\hat{\beta}_1 = 0.06538, Var(\hat{\beta}_1) = 0.0006909, \frac{\hat{\beta}_1}{s.e.(\hat{\beta}_1)} = 6.2 \Rightarrow$ Reject H_0

Three-way Contingency Tables

***R × C tables for ordinal-nominal data**

· Fit linear model 的缺點：

(1) P_j 的 fitted value 可能不介於 0 和 1 之間。

(2) $\beta_1 = P_{Z+1} - P_Z = P(Lie|Z+1) - P(Lie|Z)$

i. e. 只要年齡增加一單位，說謊的機率增加量皆相同的結果不合理。